

EXCELRATE

Data Visualization Associate Internship

Final Presentation

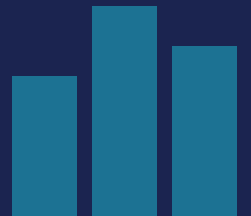
Prepared by: **Team 13**

Role: Data Visualization Associate

Week 4 Final Submission

Team PPT Screen Recording link :

<https://drive.google.com/file/d/1AsNv1JKOncS3bK1x4Q1rcT9xG9tGHM8L/view?usp=sharing>



Project Objective

The objective of this internship was to transform raw organisational datasets into meaningful business insights by performing data cleaning, SQL-based analysis, and interactive dashboard development.

OBJECTIVES ACHIEVED

- ✓ Import Opportunity and Learner datasets into PostgreSQL.
- ✓ Clean and validate raw data.
- ✓ Perform SQL analysis to generate insights.
- ✓ Integrate datasets for reporting.
- ✓ Build dashboards in Looker Studio.
- ✓ Present findings using data storytelling.

SAMPLE SQL QUERY

```
SELECT COUNT(*) AS total_opportunities  
FROM opportunity_data_raw;
```

	total_opportunities bigint
1	5999

```
SELECT COUNT(*) AS total_learners  
FROM learner_data_raw;
```

	total_learners bigint
1	15029

Datasets Overview

OPPORTUNITY DATASET

Fields

Opportunity ID • Category • Cohort • Fee
Reward • Duration • Role • Location

TOTAL RECORDS: 5,300

LEARNER DATASET

Fields

Learner Status • Payment Status • Category
Apply Date • Cohort • Reward • Completion Date

TOTAL RECORDS: 15,029

SQL QUERY USED

OPPORTUNITY CATEGORY COUNT

```
SELECT
  category,
  COUNT(*) AS total
FROM opportunity_data_raw
GROUP BY category
ORDER BY total DESC;
```

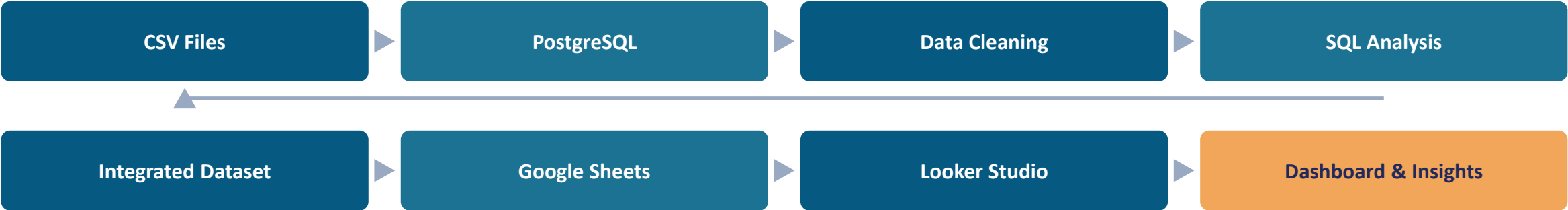
	category text	total bigint
1	Internship	1591
2	Career	1088
3	Competition	734
4	Course	632
5	Event	622

LEARNER STATUS COUNT

```
SELECT
  status,
  COUNT(*) AS learners
FROM learner_data_raw
GROUP BY status
ORDER BY learners DESC;
```

	status text	learners bigint
1	[null]	5978
2	1070	3335
3	1010	1791
4	1120	1123
5	1030	694

Project Workflow



INTEGRATION QUERY

```
SELECT *
FROM learner_data_raw l
LEFT JOIN opportunity_data_raw o
ON l.assigned_cohort = o.cohort;
```

Data Cleaning & SQL Operations

✓ Removed duplicate records

SQL

```
SELECT DISTINCT *  
FROM learner_data_raw;
```

✓ Checked missing values

SQL

```
SELECT *  
FROM learner_data_raw  
WHERE status IS NULL;
```

✓ Standardised data

SQL

```
UPDATE learner_data_raw  
SET payment_status='Unpaid'  
WHERE payment_status IS NULL;
```

SQL CONCEPTS USED

SELECT

WHERE

GROUP BY

ORDER BY

COUNT()

SUM()

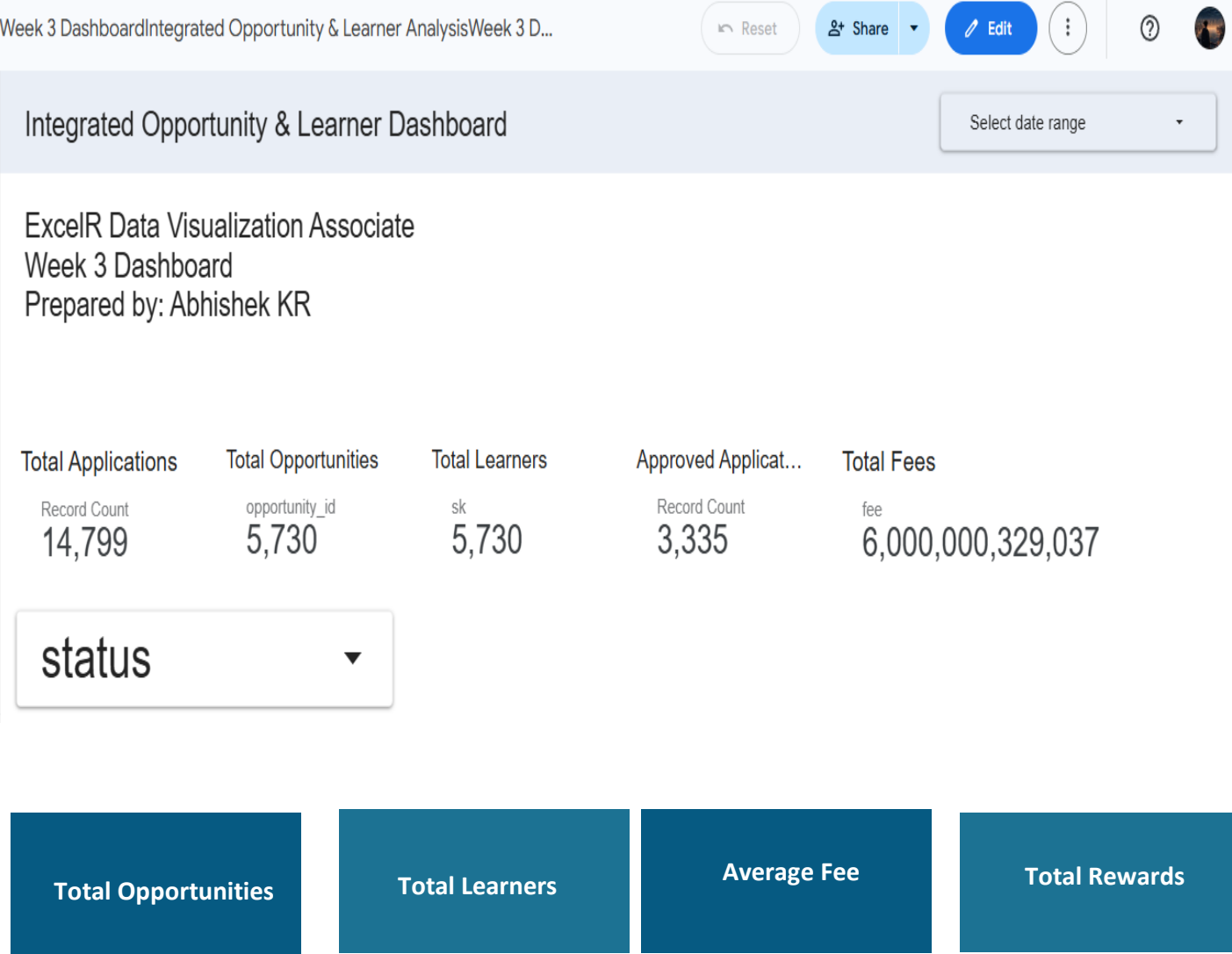
AVG()

CASE

JOIN

DISTINCT

Dashboard & KPI Cards



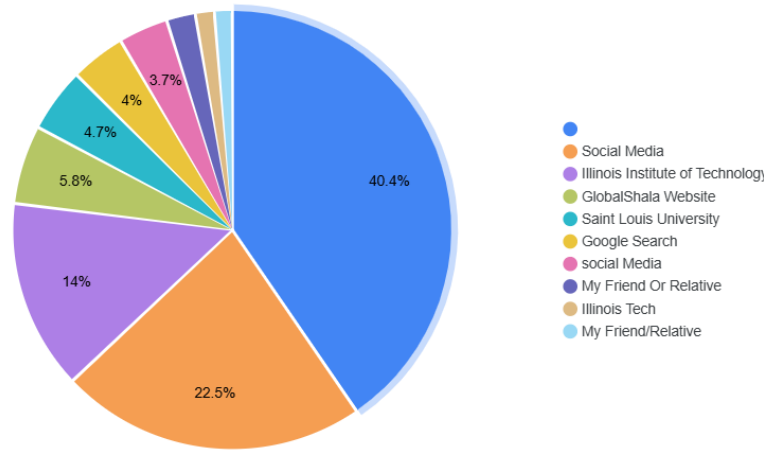
SQL QUERIES

```
SELECT AVG(fee)
FROM opportunity_data_raw;
```

```
SELECT SUM(reward)
FROM opportunity_data_raw;
```

```
SELECT COUNT(DISTINCT opportunity_id)
FROM opportunity_data_raw;
```

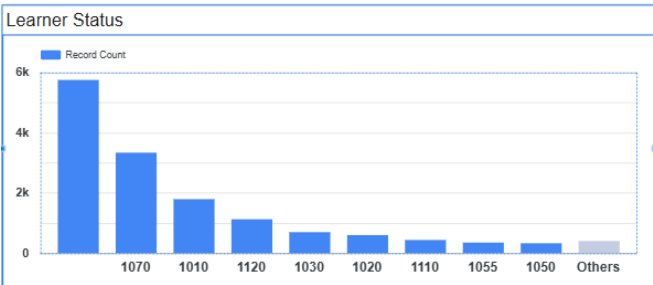
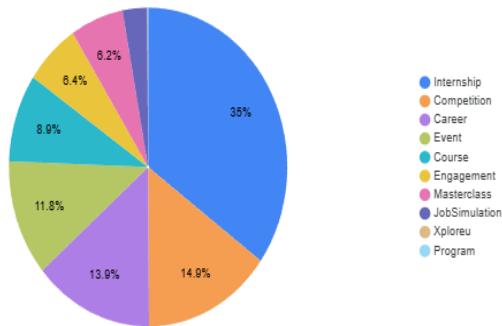
Lead Source Distribution



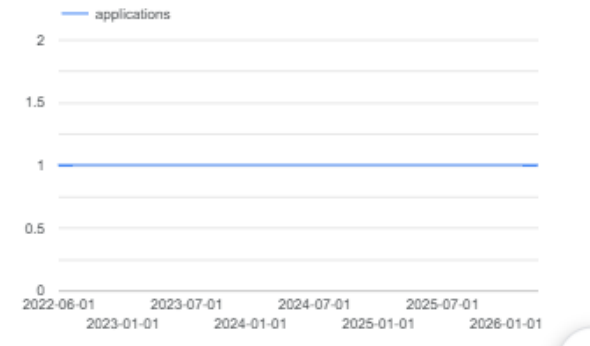
Dashboard Visualizations

Insert Dashboard Charts — visualizations include:

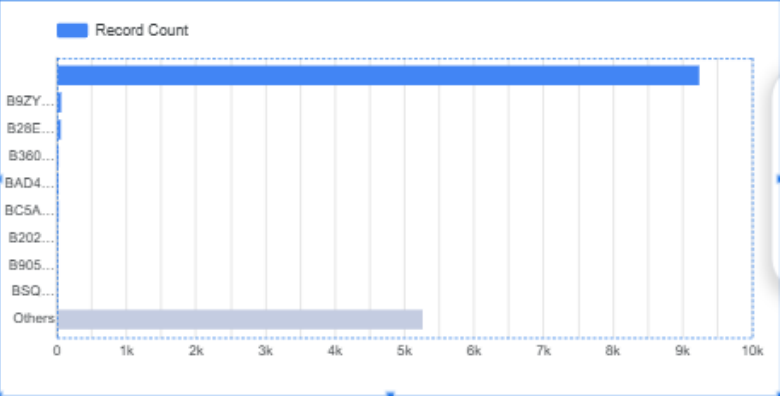
Applications by Opportunity Category



Application_Trend



Record Count by assigned_cohort



SQL EXAMPLE

```
SELECT
    payment_status,
    COUNT(*) AS total
FROM learner_data_raw
GROUP BY payment_status;
```

```
SELECT
    cohort,
    COUNT(*) AS learners
FROM learner_data_raw
GROUP BY cohort
ORDER BY learners DESC;
```

Key Findings

MAJOR INSIGHTS

- 1

Internship opportunities were the largest opportunity category.
- 2

Most learner applications belonged to a few major cohorts.
- 3

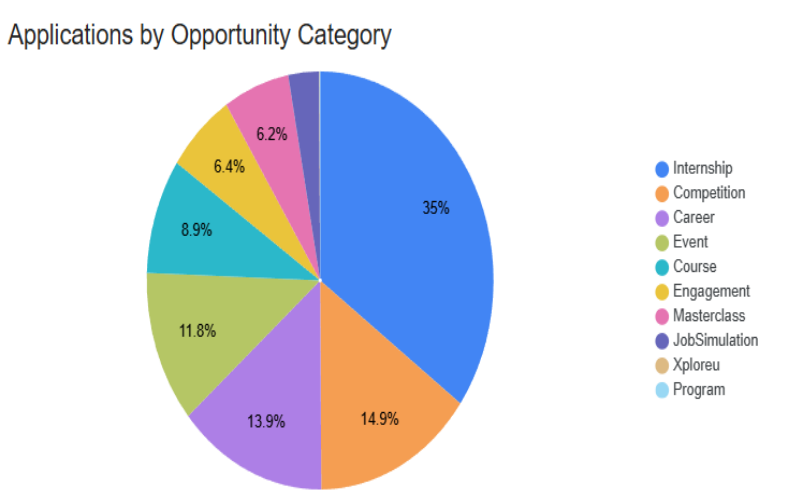
Payment completion rate can be improved.
- 4

Data cleaning significantly enhanced analysis quality.
- 5

Interactive dashboards simplified performance monitoring.

SQL USED

```
SELECT
  category,
  COUNT(*) AS total
FROM opportunity_data
GROUP BY category
ORDER BY total DESC
LIMIT 5;
```



Recommendations

- 1 Improve validation during data entry.
- 2 Reduce missing values using mandatory fields.
- 3 Automate dashboard refreshes.
- 4 Monitor payment status regularly.
- 5 Maintain consistent naming conventions.

SQL EXAMPLE

```
SELECT
  payment_status,
  COUNT(*)
FROM learner_data_raw
GROUP BY payment_status;
```

	payment_status text	count bigint
1	Unpaid	5995
2	paid	504
3	{ %22label%22: %22Zimbabwe...	1
4	unpaid	8529

Skills Gained

TECHNICAL SKILLS

PostgreSQL

SQL

Data Cleaning

Data Validation

Google Sheets

Looker Studio

Dashboard Design

Data Storytelling

PROFESSIONAL SKILLS

- Communication
- Problem Solving
- Team Collaboration
- Presentation Skills
- Analytical Thinking



A well-rounded mix of technical and professional growth over the internship.

SQL CONCEPTS MASTERED

SELECT • JOIN • GROUP BY • ORDER BY • COUNT • SUM • AVG • CASE • DISTINCT • Aggregate Functions

Challenges & Solutions

CHALLENGES

- Large datasets
- Missing values
- Inconsistent data
- Dataset integration
- Dashboard configuration

SOLUTIONS

- SQL data cleaning
- NULL value handling
- Validation using aggregate queries
- Data integration with JOIN operations
- Interactive dashboards in Looker Studio

VALIDATION QUERY

```
SELECT
  COUNT(*)
FROM learner_data
WHERE payment_status IS NULL;
```

Conclusion

INTERNSHIP OUTCOMES

- Successfully cleaned and analysed real-world datasets.
- Developed SQL-based analytical solutions.
- Built interactive Looker Studio dashboards.
- Delivered actionable business insights.
- Enhanced data visualization and storytelling skills.



Thank You

Questions?